



EyeSense

Mobile Eyesight Measurement via Time-of-Flight Phase Manipulation



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Problem & Motivation

2.2B+

People with vision impairment globally

80%

Cases preventable or treatable with early diagnosis

~0

Access to objective refraction in rural/resource-poor areas

Existing Optical Diagnostic Architecture

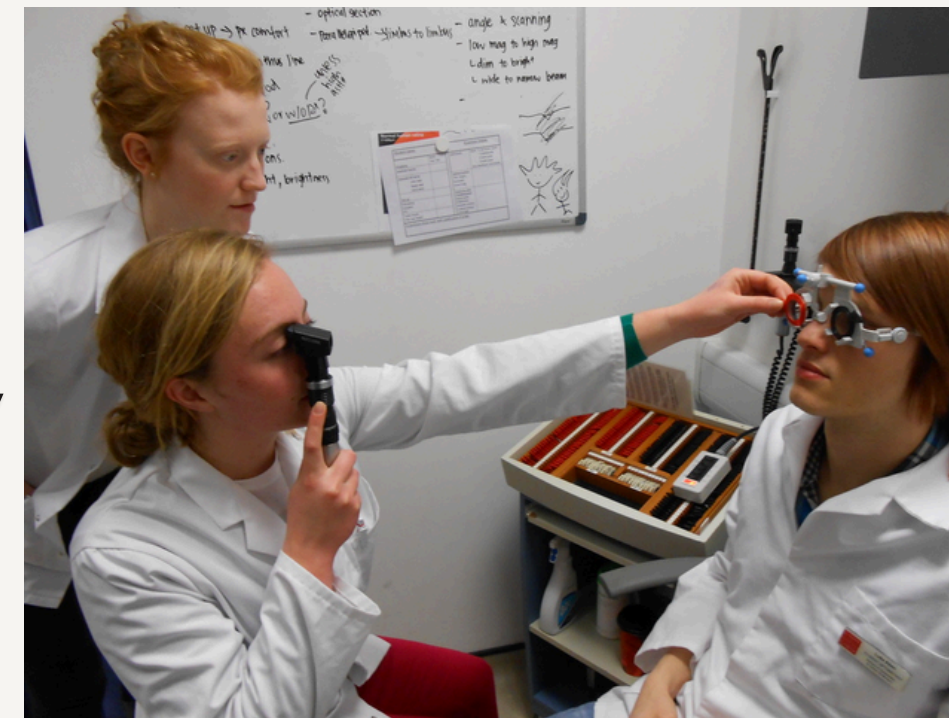
01 Autorefractors

- Objective, automated measurement of refractive error
- IR light projected onto retina
- Gold standard in clinical optometry; requires trained operator
- Cost: \$5,000–\$20,000+ per unit



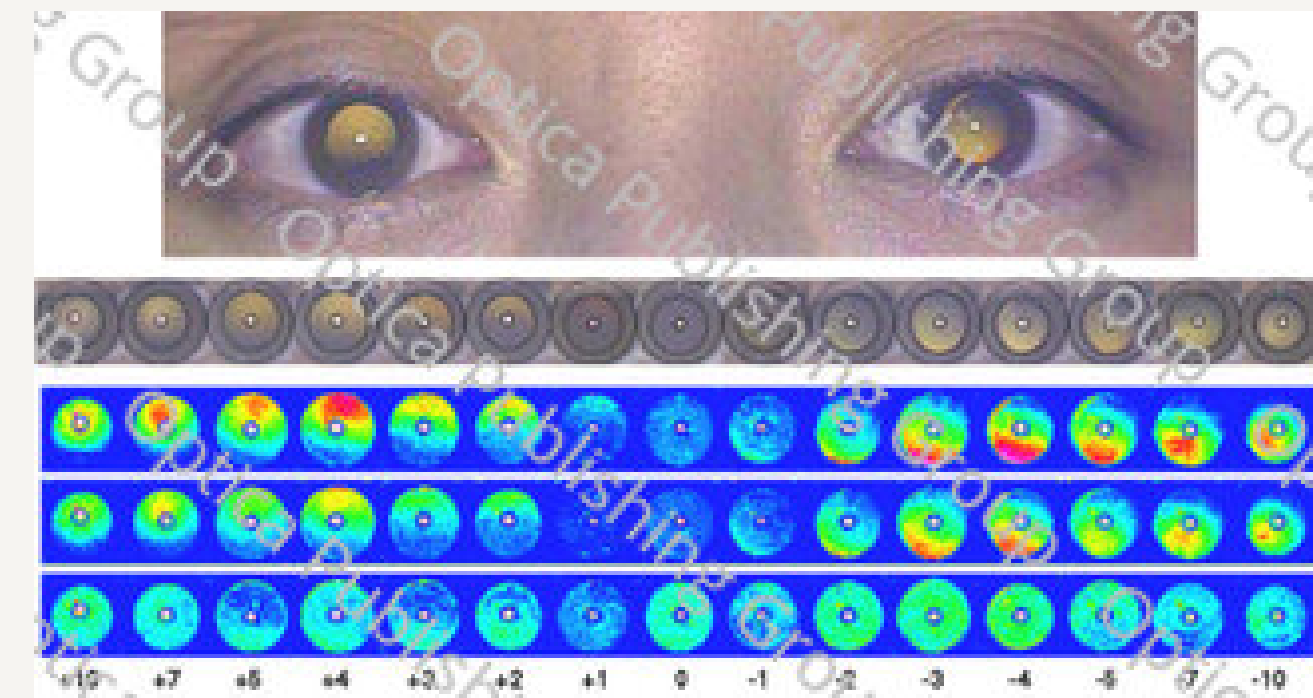
02 Retinoscopy

- Clinician shines a streak of light into the eye, observes the pupillary reflex
- Direction of reflex movement → determines refractive error
- Neutralization using trial lenses confirms the prescription



03 Eccentric Photorefraction

- Photographic / optical technique; captures the retinal reflex on camera
- Refracting eye creates a characteristic bright crescent in the pupil
- Crescent width $\propto$ magnitude of refractive error (diopters)



Related Work

Chun et al., 2020

DL + Smartphone Photorefraction

ResNet-18 on 305 visible-light eccentric photorefraction images of Korean children. 81.6% accuracy across 7 refractive classes. used Grad-CAM heatmaps to identify critical features.

Key Gap:

Visible-light flash → pupil constriction. Single ethnicity. Small dataset. No objective phase data.

Vu et al., 2024

12-LED Eccentric Photorefraction Device

Custom handheld device with 12 IR LEDs. VGG-like CNN regresses M, J0, J45 from 12-channel input. $\sigma = 0.56$ D on real data.

Key Gap:

Custom hardware not available in existing smartphones. 50k synthetic training images required. Not zero-cost for patients.

Linde et al., 2023

IR Red Reflex + EfficientNet

nun IR handheld fundus camera with dedicated IR optics. EfficientNet/Inception-v3 on 512 patients (India + NZ). 75% accuracy for spherical power.

Key Gap:

Requires dedicated \$\$ hardware (nun IR camera). No raw phase data – just IR images. Low specificity (47%). Cylinder prediction poor.

Yang et al., 2019

24-LED Custom Photorefractor

24 NIR LEDs at 850 nm, 3 meridian axes. CNN on 1216 oculi. Hough-based pupil seg + Purkinje glint removal. ± 0.9 D MSE.

Key Gap:

Requires specialized lab hardware. 1-meter fixed working distance constraint. No mobile deployment. No phase-level data.

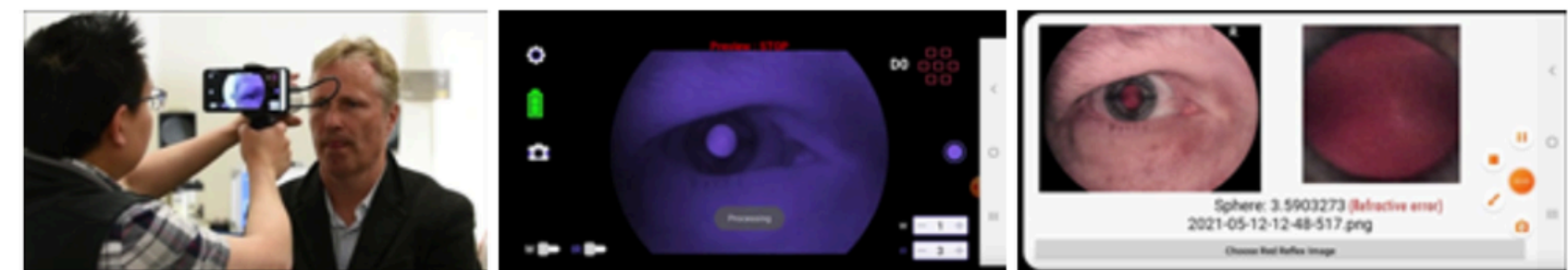


Figure 1. IR and colour red reflex image capture using nun IR.

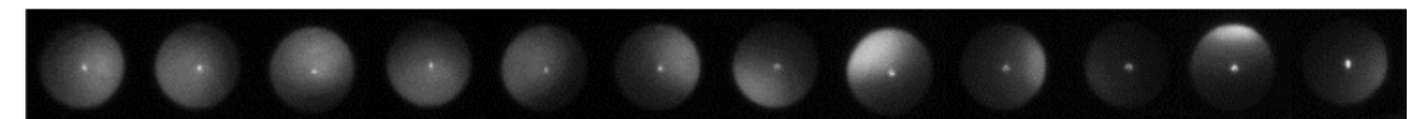


Fig. 2. Example of twelve eye pupil images captured sequentially by a different LED of the proposed device.

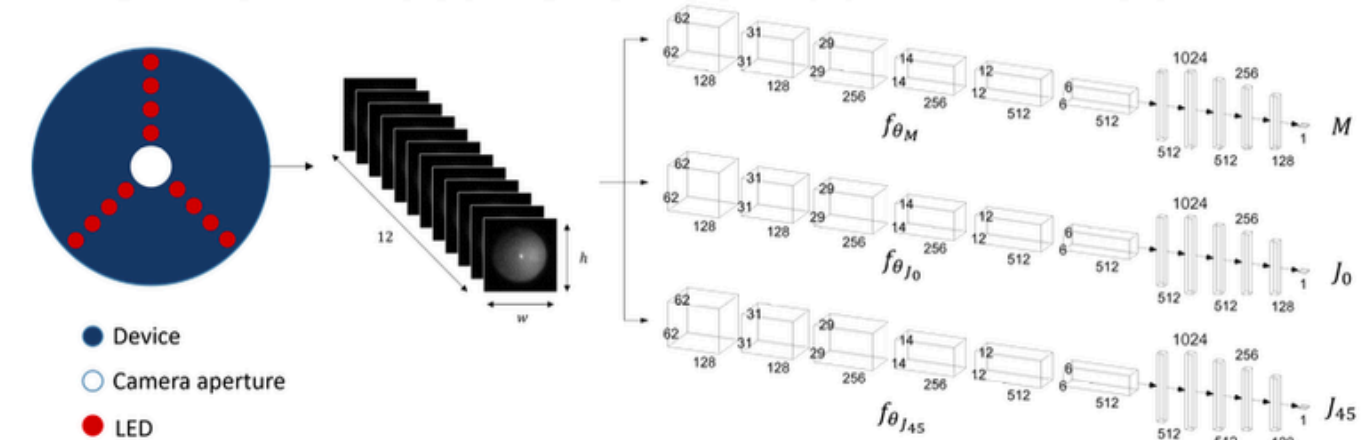
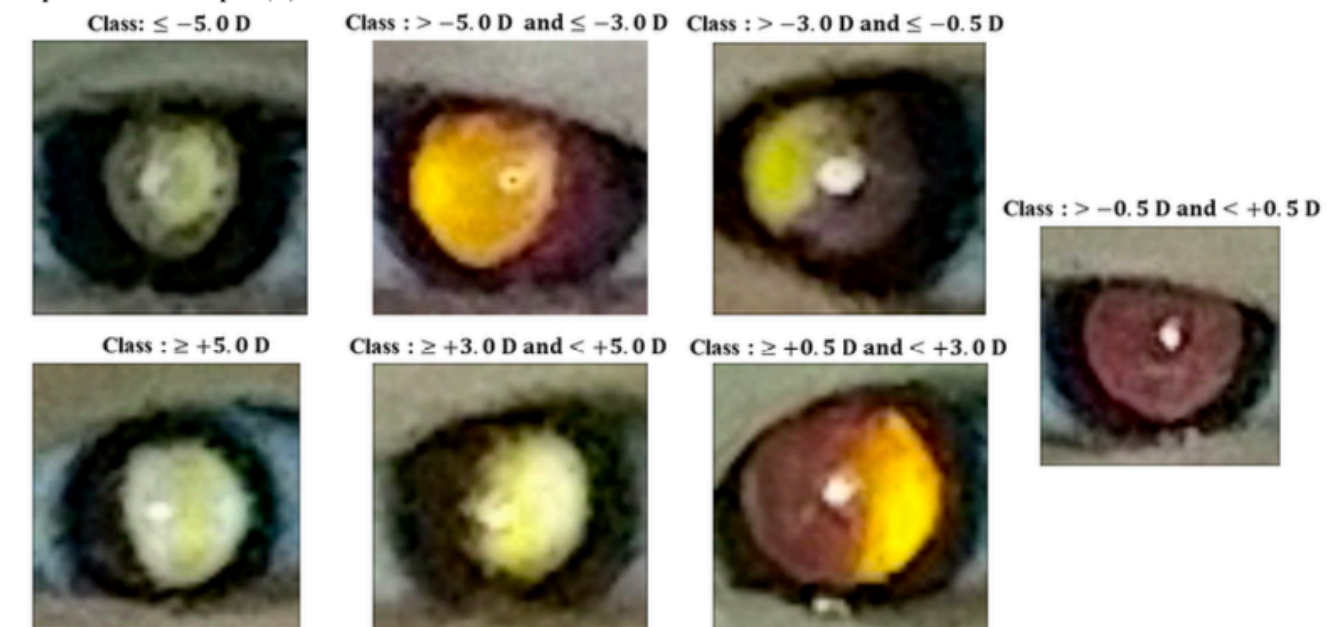


Figure 3. Examples of photorefraction images from the seven different refractor error classes. A bright crescent appears in the pupillary reflex, and its size and shape indicate the diopter (D) value.



Why Mozart

Why RGB fails — the pupil problem

- RGB cameras rely on visible light, which triggers the pupillary light reflex.

Why standard IR is better but still incomplete

- To perform IR photorefraction, they require external hardware such as LED rings or custom attachments.

Why ToF changes the physics of what is measurable

- ToF sensors measure when light returns using IR phase shifts, allowing direct depth and geometry estimation.
- This gives smartphones built-in eccentric IR illumination, enabling measurable eye-structure signals without external hardware.

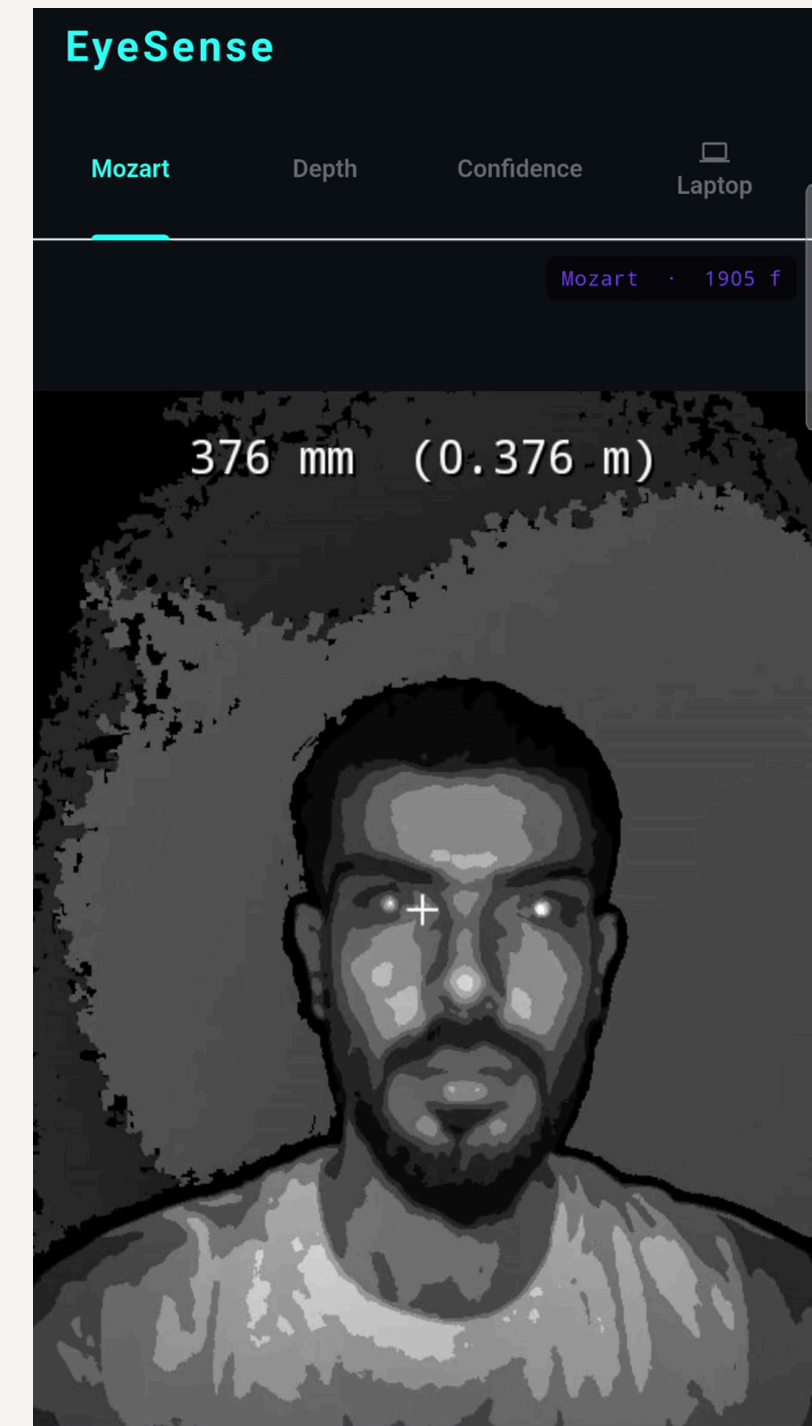
Mozart vs EyeSense: Sensor & App Modifications

Original Mozart



- Mozart's `normalizeRange()` used `RANGE_MIN = 100mm` as an absolute floor. Any pixel with depth $< 100\text{mm}$ was clamped and rendered black.
- Our fix was per-frame adaptive normalization: $\text{cutDepth} = \text{depthF}[\text{pixel}] / \text{maxDepth}$, where `maxDepth` is the maximum depth value observed in that specific frame. This means every frame normalizes to its own real content.
- The EyeSense app provides three selectable visualizations — Mozart (N2 scalar), Depth (jet colormap), and Confidence (grayscale).

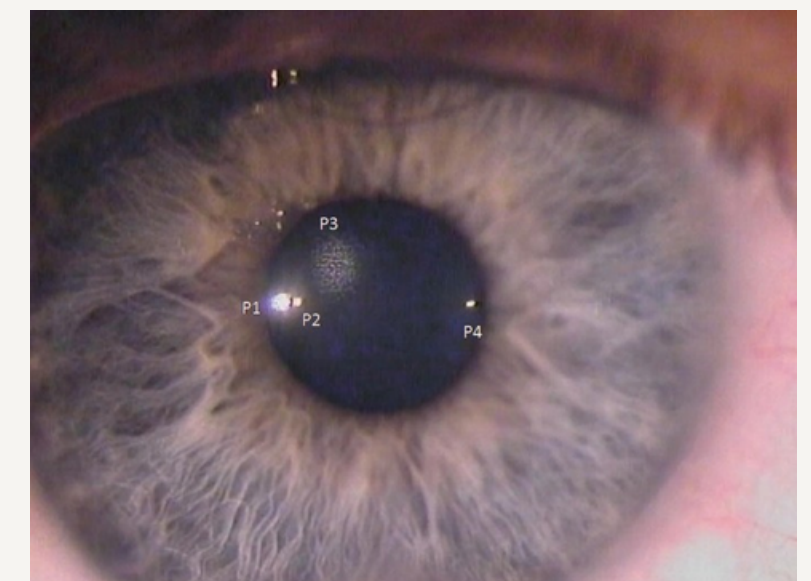
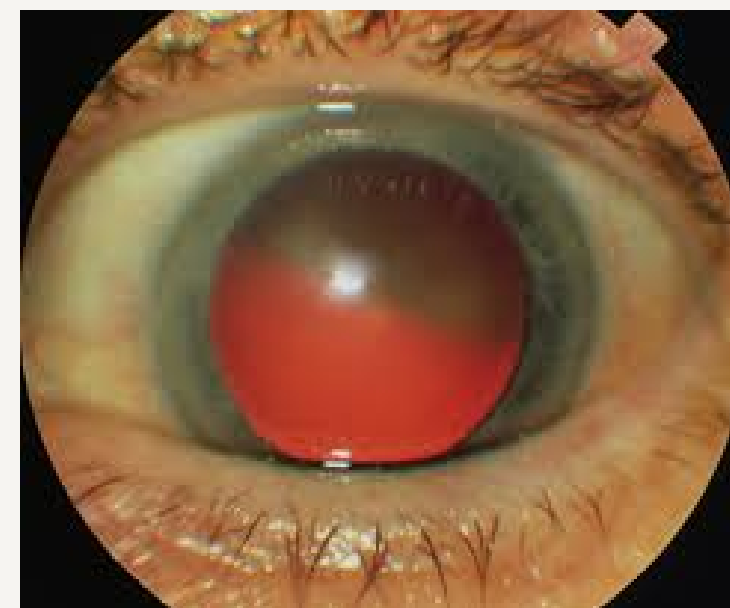
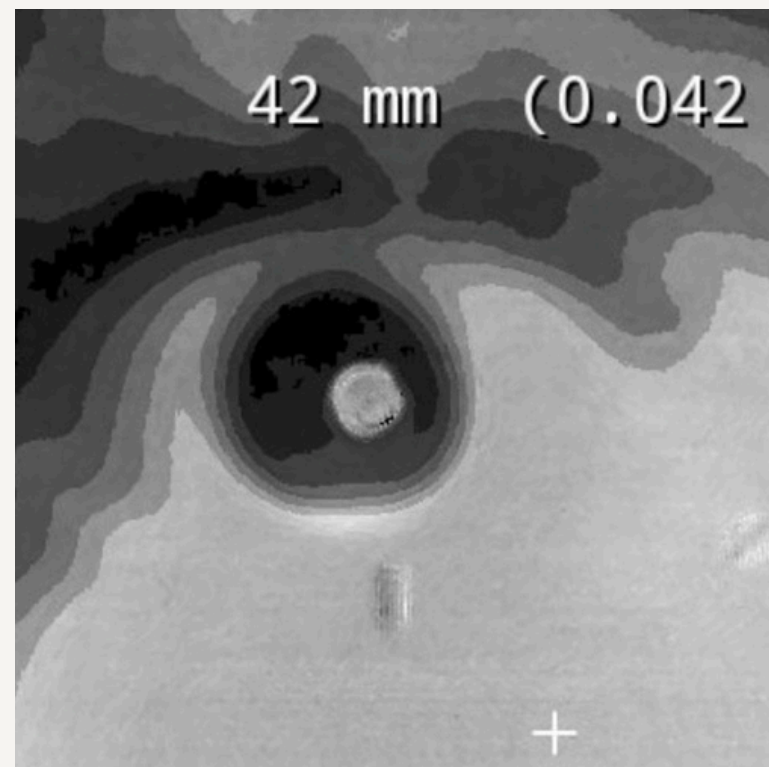
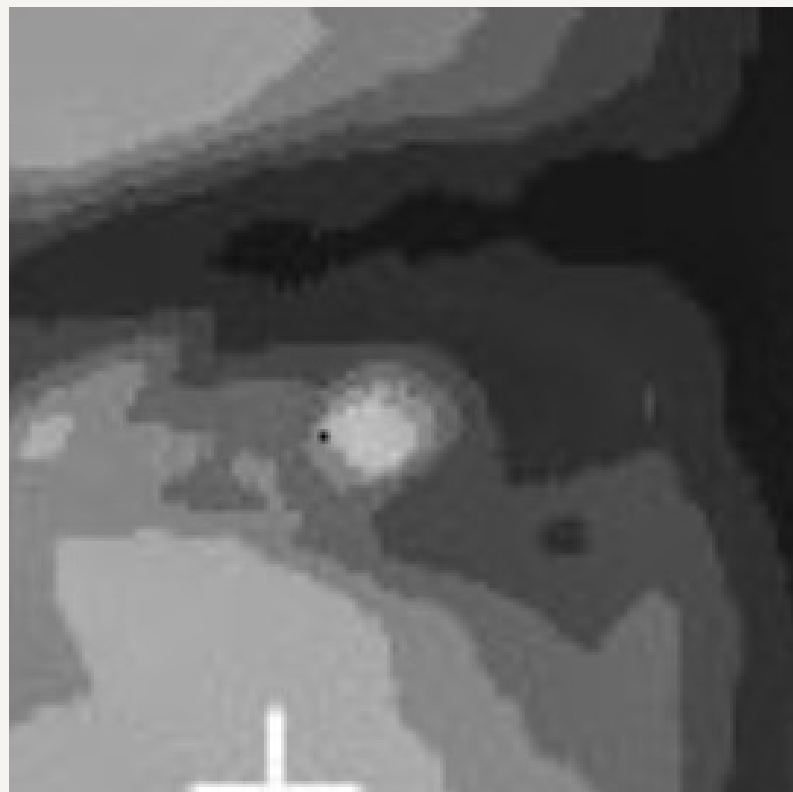
EyeSense



The Remaining Distance Limitation



- Even after all normalization fixes, we observed a consistent behavior: as the phone moves closer than approximately 20–25cm, the pupil region brightness inverts — what should appear as a bright retinal reflex starts appearing dark relative to the surrounding iris and sclera.
- Below this distance threshold, the dominant signal at the pupil is no longer the diffuse retinal reflex but rather the specular corneal reflection — the P1 Purkinje image.



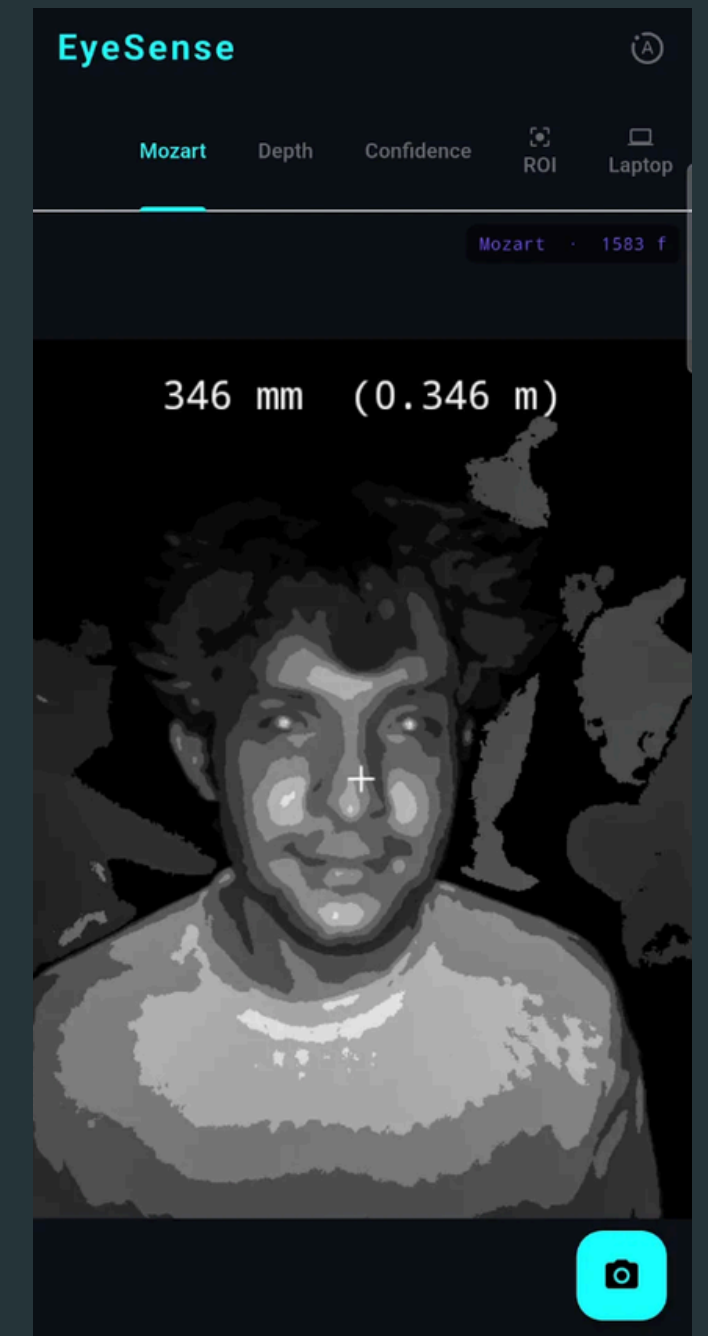
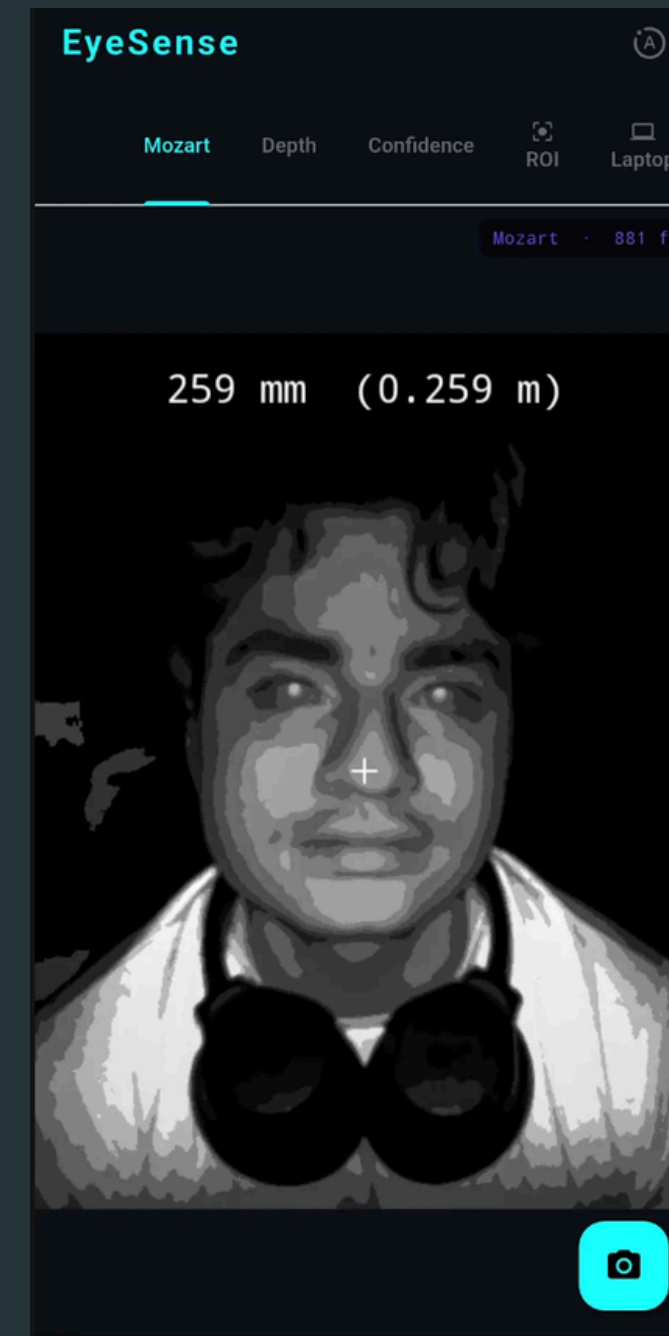
Data Collection - Phase 1

Lab Data Collection

- Conducted in IoT Lab, controlled lighting conditions
- Participants photographed using Mozart-modified ToF app on test device
- Each participant self-reported their prescription numbers (sphere)
- Data logged: Mozart phase maps + reported prescription values

Dataset Summary

- **Participants:** 80
- **Total frames captured:** 160
- **Ground Truth Type:** Self reported Rx
- **Limitation:** Ground truth unverified by clinician
- **Refractive Range (Sphere):** -9 to +2



Data Collection - Phase 2

LUMS Eye Camp (Verified Ground Truth)

setup



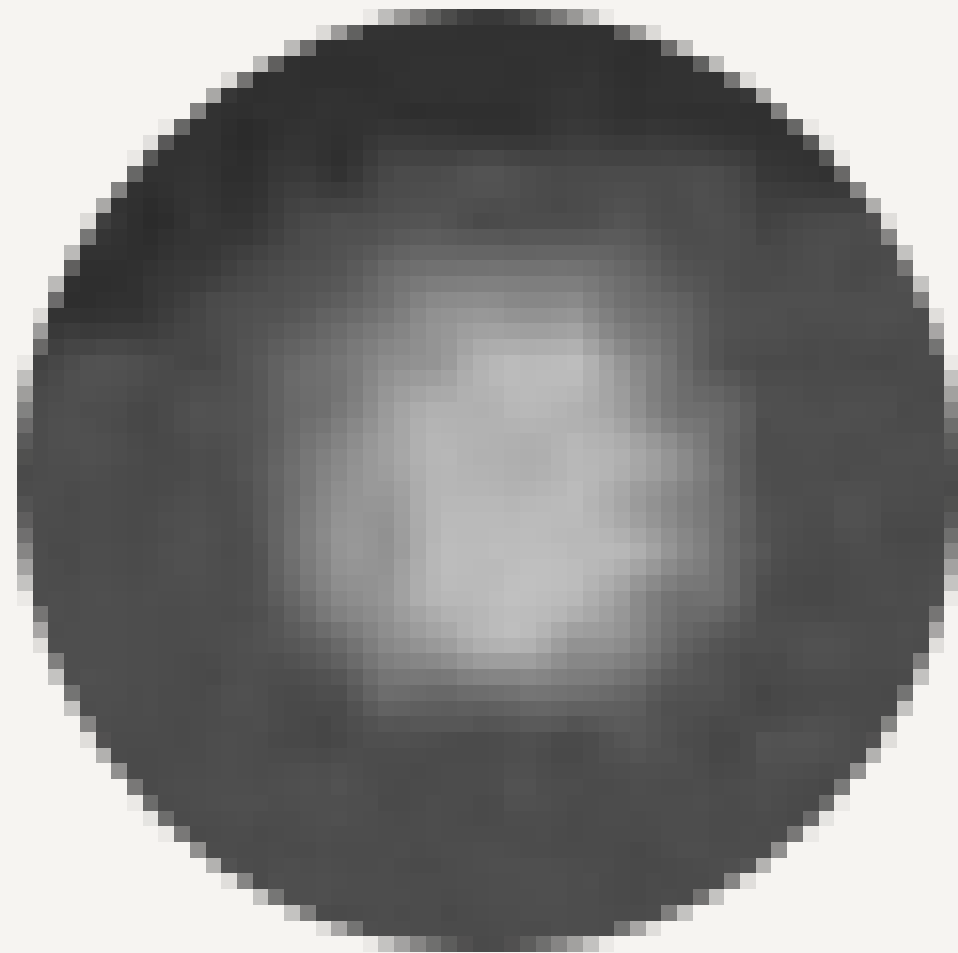
Dataset Summary

- **Participants:** 40
- **Total frames captured:** 80
- **Ground Truth Type:** Professional Autorefractor
- **Age Range:** 7 to 60
- **Refractive Range (Sphere):** -3 to +3
- **Improvement:** Eliminated self-report noise + have CYL & axis data

Labeled 160 images



3D Texture Image to Eye Prescription



Mozart texture image map

ML Pipeline



-1.5 (myopic)

Prescription number

ML Pipeline: Existing Methods

Existing Measures of Results:

- classification (accuracy %) and regression (error in diopters)
- Clinical prescription precision is $\pm 0.25\text{D}$. Acceptable screening threshold is $\pm 0.5\text{D}$ to $\pm 1.0\text{D}$

- ▶ Chun et al. 2020 — Classification, 7 refractive bins, ResNet-18, 305 images $\rightarrow$ 81.6% accuracy
- ▶ Yang et al. 2019 — Regression, custom 24-LED ring, 1216 eyes $\rightarrow \pm 0.9\text{D}$ MSE
- ▶ Vu et al. 2024 — Regression, 12-LED portable device, 50k synthetic + 3k real $\rightarrow$ 0.56D std deviation
- ▶ Linde et al. 2023 — Binary classification (myopic/not), EfficientNet, red reflex $\rightarrow$ 75% accuracy

ML Pipeline: Our Approach

Phase 1 — Ridge Regression (LOO)

Input: 8 handcrafted features per image — asymmetry, center-edge ratio, gradient, intensity stats

Conclusion: signal confirmed. Mozart maps linearly correlate with prescription. Mild myopia at 83.3%.

Ridge Regression

Phase 2 — CNN Baseline (5-fold, no augmentation)

Architecture: ResNet-18 backbone

Adam lr=0.0001, early stopping patience=20

CNN

Phase 3 — Augmentation Ablation

Augmentations tested: horizontal flip, color jitter ± 0.3 , $\pm 10^\circ$ rotation, Gaussian blur

Result: MAE 1.26D — worse than no augmentation

Why: horizontal flip inverts left-right asymmetry, a physically directional signal in Mozart maps

CNN +
Augmentation

Phase 4 — Ensemble

Ridge and CNN fail on different images — their errors are partially uncorrelated

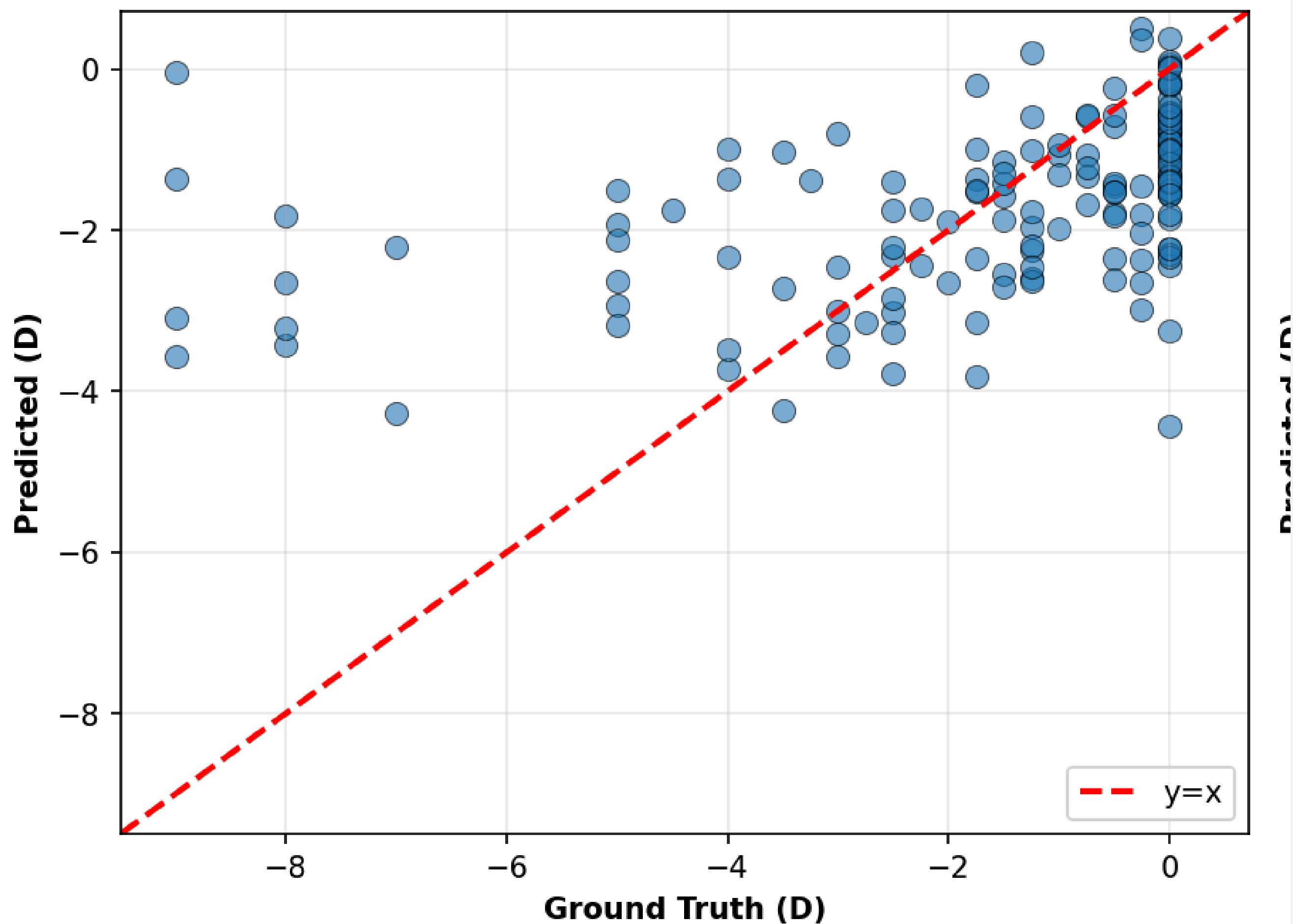
Ensemble cancels specific failure modes of each model

0.4/0.6 weighting: CNN carries more weight but Ridge stabilizes tail predictions

Ensemble
(Ridge + CNN
Baseline)

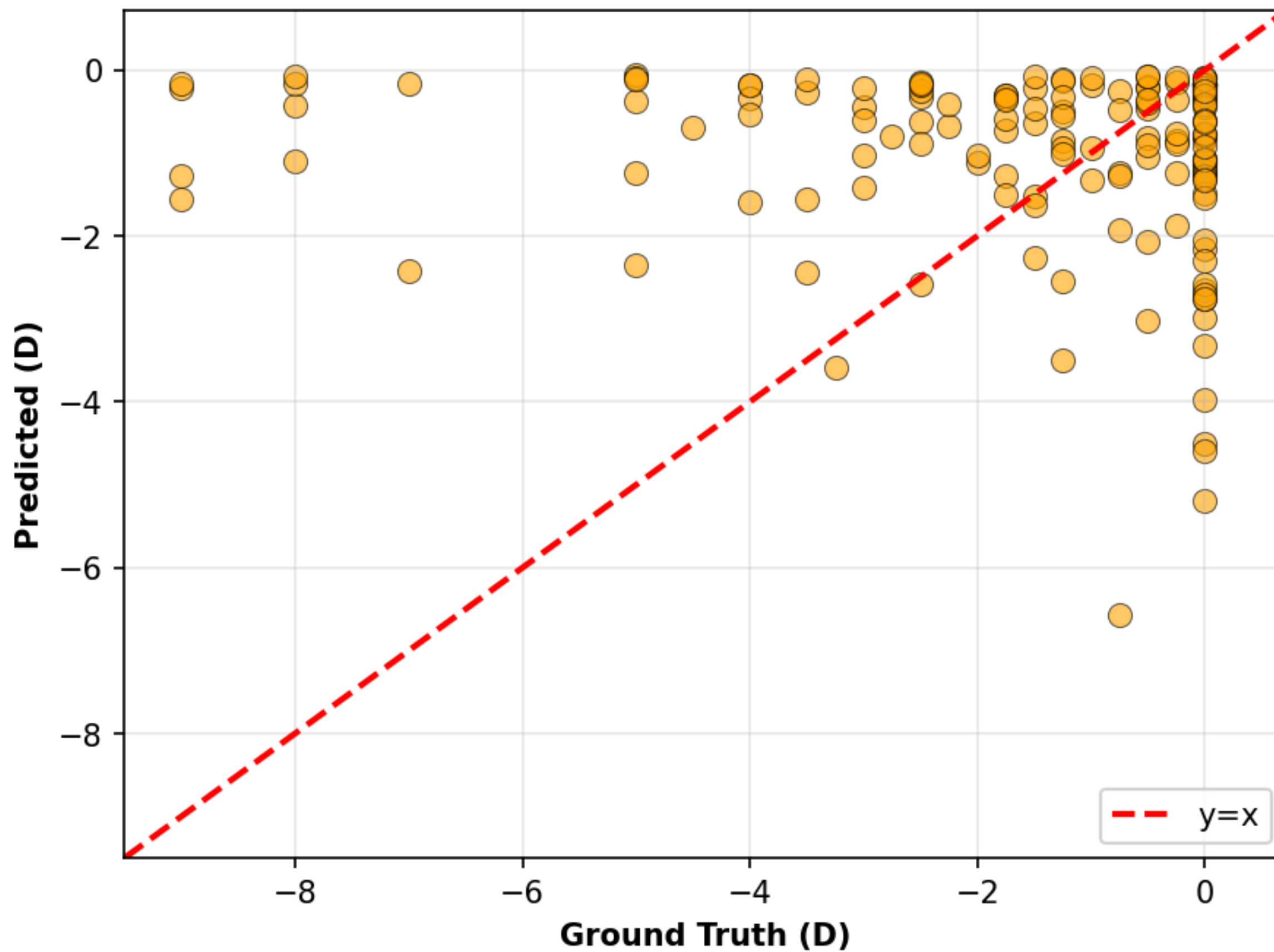
Ridge Regression

MAE = 1.3696D

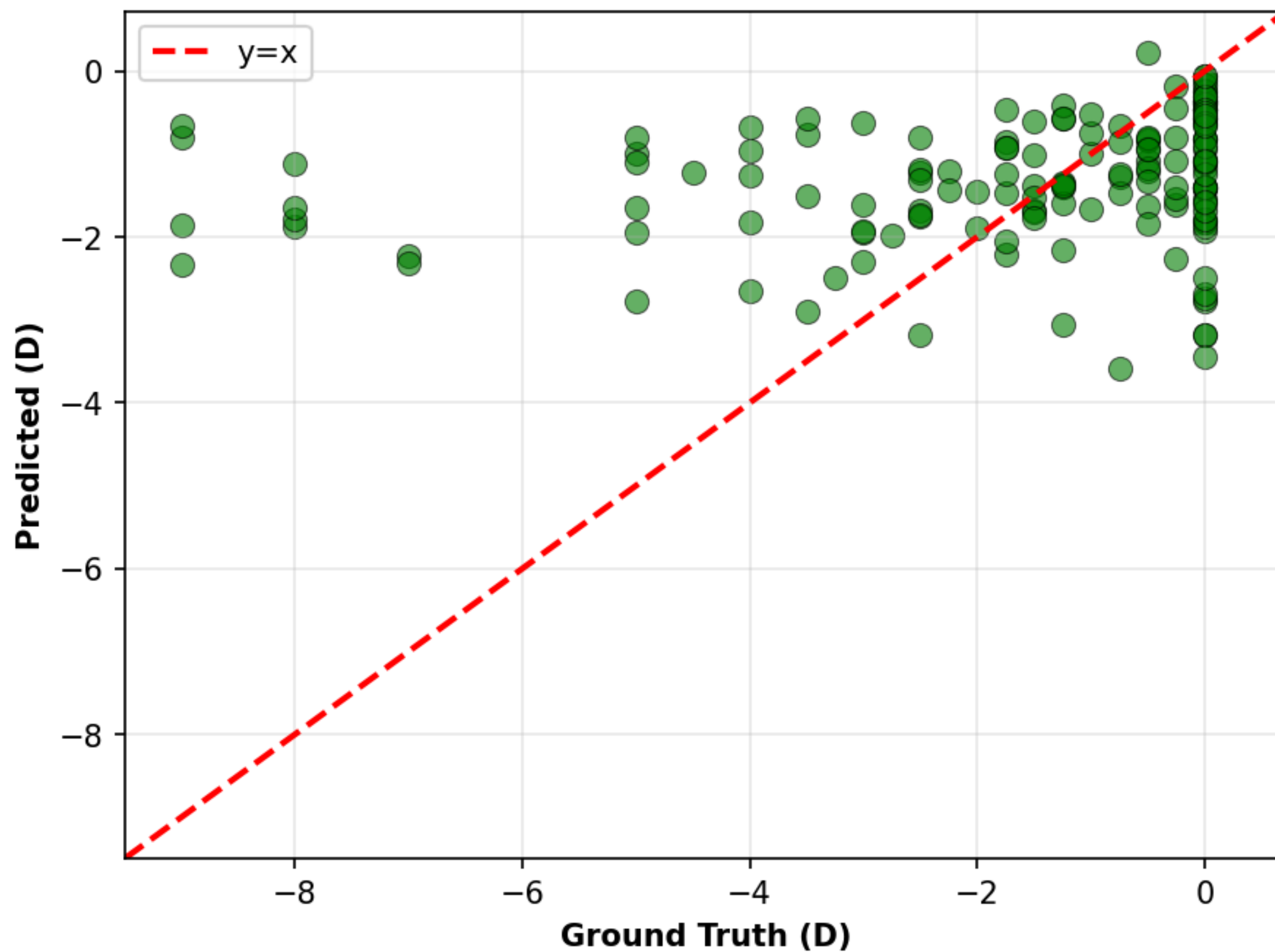


CNN Regression

MAE = 1.7694D



ENSEMBLE (Average)
MAE = 1.4449D



Model Performance Summary

Model	MAE (D)	RMSE (D)	Acc ±1.0D	Acc ±1.5D
Ridge (LOO)	1.2860	1.8976	53.8%	70.6%
CNN Baseline	1.1860	1.9503	63.1%	75.6%
CNN Augmented	1.2586	2.0412	60.6%	70.0%
Ensemble (Ridge+CNN)	1.1506	1.8648	64.4%	76.2%

Future Direction

01 Direct Crescent Width Measurement (N1) — The Geometric Approach

- Move from gradient-based inference → direct geometric measurement of crescent width
- All variables measurable directly from ToF phase data
- Eliminates CNN as a black box for the core refraction calculation

02 Direct N2 — Astigmatism Axis from Crescent Orientation

- Goal: precise angular measurement of crescent tilt → direct cylinder and axis values
- Full prescription output: sphere + cylinder + axis — without patient input

03 Dataset Expansion and Model Generalization

- Target: 1,000+ participants across age groups, iris colorso4233
- Pediatric cohort especially needed — primary use case is childhood screening

Thank
You

